Quantitative Macro Theory **HW #1**
Spring 2019 **Due at 10pm on Friday 1/25/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code (it can be in MATLAB, C, Fortran, Python, Julia, anything you want). Be prepared to explain your code in class. Email me that pdf file before the deadline.

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1. (50 points) Consider a deterministic version of the neoclassical growth model:

$$\max_{l_t, c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t \frac{\left(c_t - \frac{l_t^2}{2}\right)^{1-\sigma}}{(1-\sigma)},$$

subject to,

$$c_t + k_{t+1} = A k_t^{\theta} l_t^{1-\theta} + k_t(1-\delta).$$

   (a) (10 points) Write down the Euler equation and the FOC for labor, l_t. Note that now you have a system of 2 equations. Set $\sigma = 1.5$, $\beta = 0.95$, $\theta = 0.3$, $\delta = 0.1$, and

$$A = \left(\frac{1}{\theta \beta (1-\theta)^{\frac{1-\theta}{1+\theta}}} \right)^{\frac{1+\theta}{2}}.$$

   What are the steady state value of capital (k_{ss}) and labor (l_{ss})?

   (b) (10 points) Set $k_0 = 0.85$. What is the value of l_0? Now imagine you know that $k_2 = 0.95$. Find the optimal values of k_1 and l_1 using bisection. Make a table reporting each iteration of bisection until you found the solution. The table should also report the solution and how many seconds it took. Use other method of your choice to solve the Euler equation and report the same things.

   (c) (20 points) Starting at $k_0 = 0.75$ find the path of k_t and l_t from $t = 0$ until $t = 200$ using the extended path method I described in class. Starting from your guess $\{k_t^0\}$, you will have a revised path for $\{k_t\}$ in each iteration j, i.e., $\{k_t^j\}$. Plot the paths for capital obtained at $j = \{0, 1, 2, 5, 10\}$ together with the optimal path. Do the same for l_t.

   (d) (10 points) Plot the paths for output, employment, investment, consumption, and the real interest rate from $t = 0$ until $t = 200$.

2. (50 points) Let's use the same model that in the previous question. But instead of constant productivity, consider

$$A_t = A \times (1 + 0.05 \times (\cos(\pi \times t/25) - 1)) \text{ for } t = 0, ..., 50.$$

and $A_t = A$ for $t = 51, ..., 200$.

(a) (10 points) Starting at $k_0 = k^*$, find the optimal path of k_t and l_t from $t = 0$ until $t = 250$ using the extended path method I described in class. Plot the paths for k_t and l_t that you obtained in iteration $j = \{0, 1, 2, 5, 10\}$ together with the optimal path.

(b) (10 points) Imagine the previous solution represents a recession that surprised the economy in steady state and lasted for 25 months (about 2 years) while productivity was decreasing. Plot and explain the behavior of output, employment, investment, consumption, and the real interest rate from $t = 1$ to $t = 25$.

(c) (30 points) Obtain quarterly data for the US on employment, investment, consumption, and the real interest rate and compare them with what you obtained from the model. To compare it with the model, start in the third quarter of 2007 until the second quarter of 2009 (8 quarters). What is the greatest success of the model in term of matching the data? What about the greatest failure? (Note that you may want to detrend some of the series. Report what you did.) As a reference for the behavior of key time series during the recession, look at Figure 4 in this paper "Understanding the Great Recession," by Christiano, Eichenbaum, Trabandt. AEJ: Macro, Jan 2015. Feel free to read all the paper to learn more about the Great Recession.